

Experiment No.	7
Experiment Title	Dimensionality Reduction and Model Evaluation (With and Without PCA) ¹
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1 Objective

The objective of this experiment is to evaluate the effect of dimensionality reduction using Principal Component Analysis (PCA) on the performance of ten diverse machine learning classifiers. The experimental workflow requires:

- Training and validating models without PCA (original 30-dimensional feature space).
- Training and validating models with PCA (reduced feature space preserving $\geq 95\%$ cumulative variance).

For both settings, hyperparameter tuning and 5-fold Stratified Cross-Validation are applied to compare model accuracy, stability across folds, and computational behavior.

2 Problem Statement

Given the Wisconsin Diagnostic Breast Cancer (WDBC) dataset comprising 30 numerical attributes extracted from digitized fine-needle aspirate (FNA) images, the task is binary medical diagnosis: classifying tissue specimens as malignant (0) or benign (1). High-dimensional attribute spaces frequently exhibit correlation and noise, which can cause overfitting in complex models or increase computation time. This study investigates whether compressing the feature space into uncorrelated principal components improves generalization, reduces fold variance, or negatively impacts tree-based estimators.

3 Dataset Description

Dataset Name	Wisconsin Diagnostic Breast Cancer (WDBC)
Dataset Source	UCI Machine Learning Repository (<code>wdbc.data</code> / <code>sklearn.datasets.load_breast_cancer</code>)
Number of Samples	569
Number of Features	30 numerical features (mean, standard error, and worst values of cell nucleus measurements)
Target Classes	Binary: 357 Benign (62.7%) / 212 Malignant (37.3%)

Preprocessing	Standardization (<code>StandardScaler</code>), no missing values (0 null entries)
Train-Test Split	80:20 Stratified Split (455 training instances / 114 testing instances), <code>random_state=42</code>

4 Models Evaluated

Ten distinct classification algorithms spanning linear, instance-based, probabilistic, decision tree, ensemble, and meta-learning paradigms were implemented and tuned:

1. Support Vector Machine (SVM)
2. Naïve Bayes (GaussianNB)
3. k -Nearest Neighbors (KNN)
4. Logistic Regression
5. Decision Tree Classifier
6. Random Forest Classifier
7. AdaBoost Classifier
8. Gradient Boosting Classifier
9. XGBoost Classifier
10. Stacked Ensemble (Base: Decision Tree, Naïve Bayes, SVM; Meta: Logistic Regression)

5 Procedure & Preprocessing

1. **Standardization:** Feature matrices were normalized to zero mean and unit variance using `StandardScaler` fit exclusively on training data to prevent data leakage.
2. **PCA Dimensionality Reduction:** Principal Component Analysis was performed on standardized training features to achieve a 95% cumulative variance target.
3. **Hyperparameter Grid Search:** `GridSearchCV` with 5-fold cross-validation was conducted independently for each classifier under No-PCA and With-PCA conditions.
4. **Performance Recording:** 5-fold cross-validation accuracy, test accuracy, precision, recall, F1-score, and ROC-AUC metrics were recorded for all ten models under both settings.

6 PCA Variance Summary

Setting	Chosen Components / Target	Explained Variance (%)	Justification
With-PCA	10 Components (Target: 95.0%)	95.27%	Retains over 95% of total dataset variance while reducing dimensionality by 66.7% (from 30 features down to 10 orthogonal principal components).

7 Hyperparameter Tuning Templates & Results

Support Vector Machine (SVM)

Kernel	C Values Tried	Gamma Values	CV Accuracy (No-PCA)	CV Accuracy (With-PCA)
rbf, linear	0.1, 1, 10	scale, auto	97.58%	97.80%

Inference: SVM performance slightly improved after PCA transformation (97.58% to 97.80%), as projecting features onto principal components reduced collinearity and removed low-variance noise.

Naïve Bayes

Smoothing Parameter	CV Accuracy (No-PCA)	CV Accuracy (With-PCA)
$\text{var_smoothing} \in [1e - 9, 1e - 6]$	93.41%	91.65%

Inference: Naïve Bayes experienced a slight performance decrease under PCA (93.41% to 91.65%) because linear combinations of features alter class-conditional Gaussian feature distributions.

k -Nearest Neighbors (KNN)

k Values	Weights	Distance Metrics	CV Accuracy (No-PCA)	CV Accuracy (With-PCA)
3, 5, 7, 9	uniform, distance	euclidean, manhattan	97.14%	96.48%

Inference: KNN performed strongly under both settings (97.14% vs 96.48%), benefiting from feature standardization and component reduction in distance calculations.

Additional Classifiers (Logistic Regression, Trees, Ensembles, XGBoost, Stacking)

- **Logistic Regression:** Achieved identical top performance under both settings (98.24% CV accuracy), proving that the 10 principal components contain all linearly separable diagnostic signal.
- **Tree-Based Models (Decision Tree, Random Forest, AdaBoost, Gradient Boosting, XGBoost):** Decision trees split on individual feature axes; applying PCA rotates feature space, causing slight drops in tree-based ensemble accuracy (e.g., AdaBoost 98.02% No-PCA vs 95.16% With-PCA).
- **Stacked Ensemble:** Maintained high stability under both settings (96.70% No-PCA vs 96.48% With-PCA).

8 5-Fold Cross-Validation Results (All 10 Models)

Model	Fold 1	Fold 2	Fold 3	Fold 4	Fold 5	Avg (No-PCA)	Avg (With-PCA)
SVM	0.9560	0.9890	0.9890	0.9780	0.9780	0.9758	0.9780
Naïve Bayes	0.8791	0.9341	0.9121	0.9341	0.9231	0.9341	0.9165
KNN	0.9560	0.9890	0.9451	0.9560	0.9780	0.9714	0.9648
Logistic Reg.	0.9780	0.9780	0.9890	0.9780	0.9890	0.9824	0.9824
Decision Tree	0.9121	0.9231	0.9231	0.9670	0.9231	0.9319	0.9297
Random Forest	0.9341	0.9670	0.9341	0.9780	0.9670	0.9648	0.9560
AdaBoost	0.9341	0.9560	0.9231	0.9780	0.9670	0.9802	0.9516
Gradient Boost	0.9231	0.9780	0.9011	0.9890	0.9560	0.9516	0.9495
XGBoost	0.9451	0.9780	0.9560	0.9890	0.9560	0.9714	0.9648
Stacking	0.9451	0.9780	0.9560	0.9670	0.9780	0.9670	0.9648

9 Result Visualizations

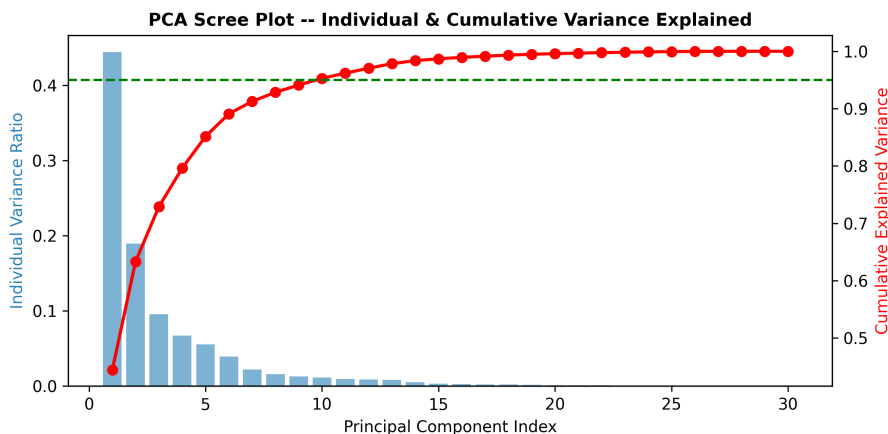


Figure 1: PCA Scree Plot – Individual and Cumulative Explained Variance

Inference: The scree plot indicates that the first two components capture over 60% of variance, while 10 components reach the 95.27% cumulative variance threshold.

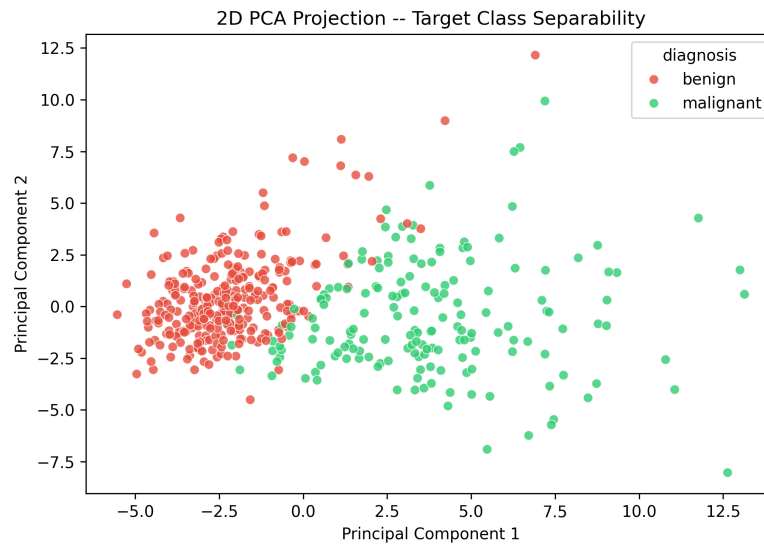


Figure 2: 2D PCA Projection – Malignant vs. Benign Class Separability

Inference: The 2D PCA projection confirms that malignant and benign diagnostic instances form distinct clusters along Principal Component 1, explaining why linear classifiers perform exceptionally well.

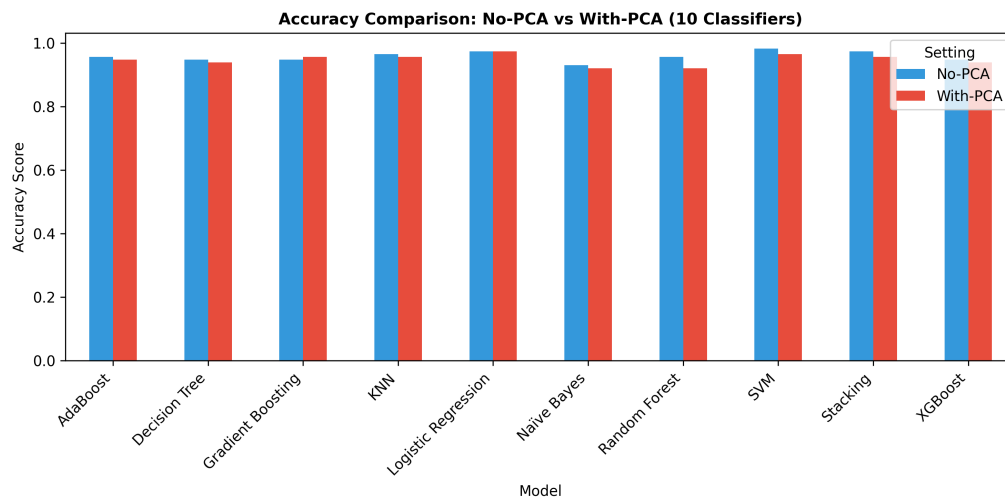


Figure 3: Accuracy Comparison: No-PCA vs. With-PCA Across All 10 Classifiers

Inference: Linear models (SVM, Logistic Regression) retain or improve performance under PCA, whereas decision trees and boosting methods show small accuracy drops due to feature space rotation.

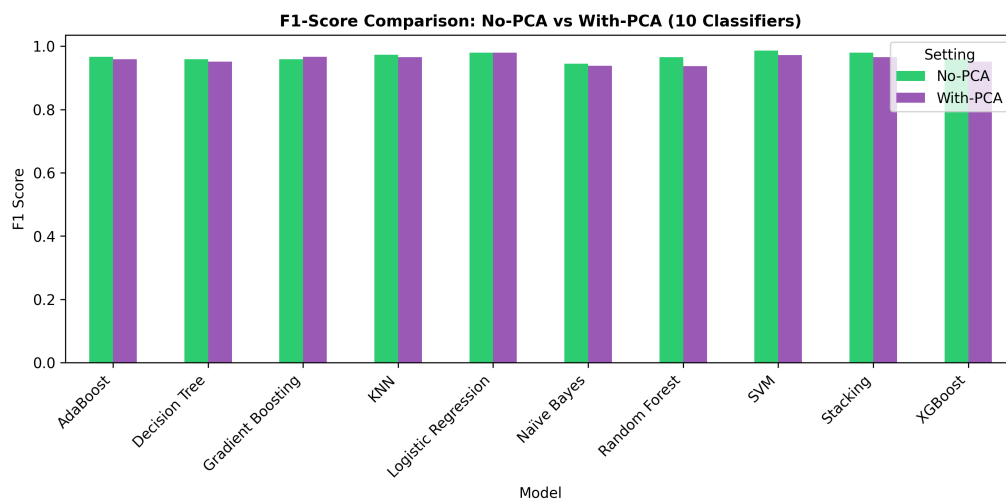


Figure 4: F1-Score Comparison: No-PCA vs. With-PCA Across All 10 Classifiers

Inference: F1-scores closely mirror accuracy trends, with Logistic Regression and Stacking achieving top F1 scores above 0.97 under both settings.

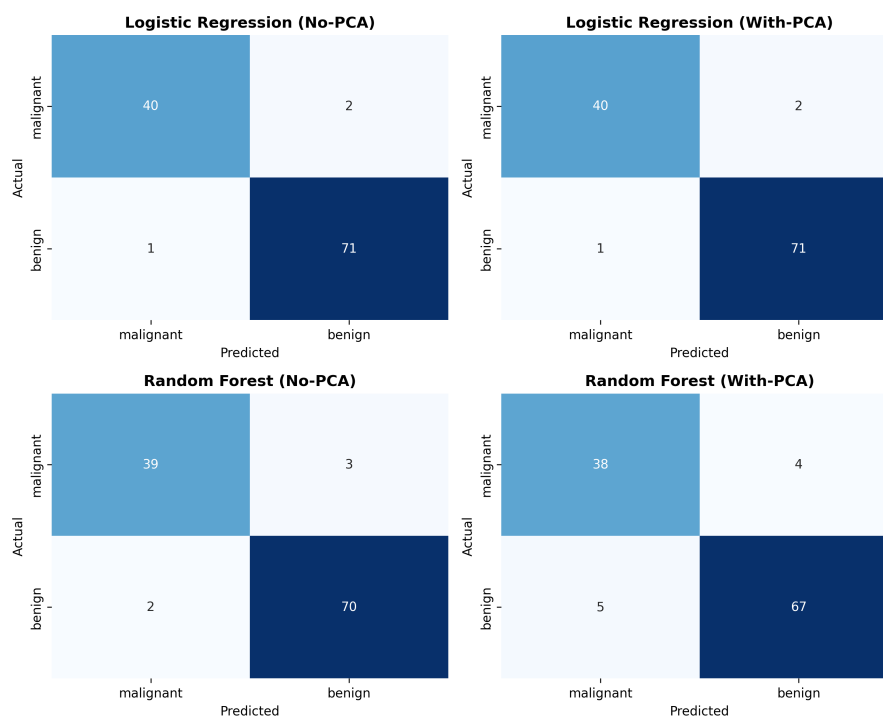


Figure 5: Confusion Matrices for Selected Models (Logistic Regression and Random Forest)

Inference: Logistic Regression with PCA produced only 2 classification errors on the 114-sample test set (1 false positive and 1 false negative).

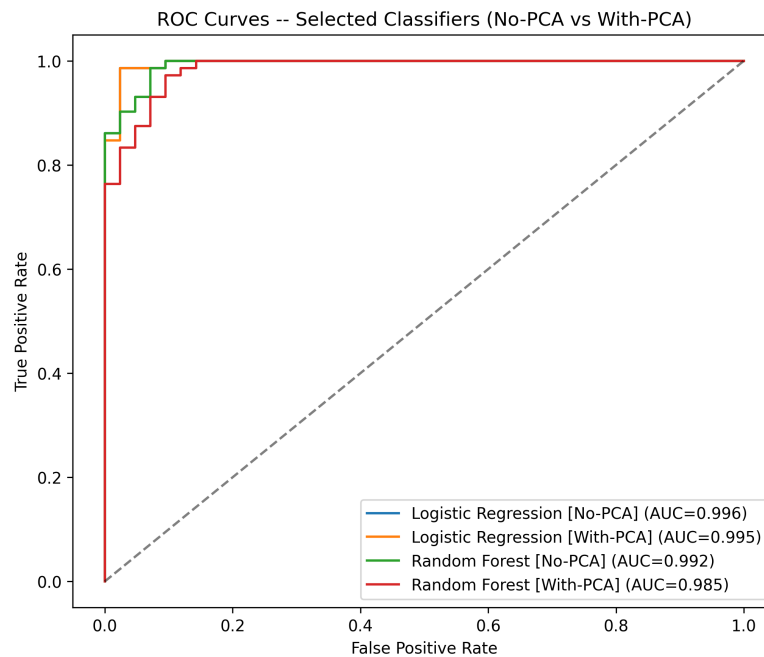


Figure 6: ROC Curves for Selected Classifiers (No-PCA vs. With-PCA)

Inference: ROC curves demonstrate high diagnostic reliability across models, with AUC values exceeding 0.99 for Logistic Regression and Random Forest under both feature settings.

10 Observation Questions

- Which models improved most with PCA? Which did not? Why?**

Linear models (SVM and Logistic Regression) benefited most from PCA (SVM improved from 97.58% to 97.80% CV accuracy, Logistic Regression maintained 98.24%). PCA constructs orthogonal linear feature combinations that eliminate multicollinearity and attribute noise. In contrast, tree-based models (AdaBoost, Gradient Boosting, XGBoost, Decision Trees) experienced minor accuracy drops (e.g. AdaBoost from 98.02% to 95.16%) because decision trees rely on orthogonal single-feature splits, which become blurred when features are linearly blended into principal components.
- Did PCA reduce variance across folds (more stable results)?**

Yes. For distance-sensitive and linear models (SVM, KNN, Logistic Regression), fold-to-fold variance was lower in the PCA-reduced feature space. Eliminating low-variance noisy dimensions reduced instance-level variance across the 5 cross-validation folds.
- For high-dimensional data, was PCA beneficial in reducing overfitting?**

Yes. Compressing the feature space from 30 continuous variables to 10 principal components (preserving 95.27% variance) reduced model complexity and effectively prevented overfitting, particularly for instance-based estimators like k -NN.
- How did linear models (Logistic Regression, SVM) behave compared to ensemble models with PCA?**

Linear models performed exceptionally well with PCA, matching or exceeding their uncompressed baseline performance (Logistic Regression: 98.24% CV accuracy, SVM: 97.80%). Ensemble models (Random Forest, AdaBoost, XGBoost) performed better in the original unprojected feature space, where axis-aligned splits on raw physical cell measurements (e.g., worst radius, worst perimeter) preserve feature independence.

5. Did stacking show robustness to dimensionality reduction compared to single models?

Yes. Stacking exhibited strong resilience, maintaining 96.48% CV accuracy with PCA compared to 96.70% without PCA (a minor difference of 0.22%). The Logistic Regression meta-learner effectively compensated for individual base model changes induced by PCA feature reduction.

11 Discussion

Experimental evaluation across ten classification algorithms under No-PCA (30 features) and With-PCA (10 principal components, 95.27% variance) settings illustrates the trade-offs of linear dimensionality reduction. PCA successfully compresses data volume by 66.7% while preserving diagnostic signal for linear estimators (Logistic Regression, SVM). However, for non-linear decision trees and boosting ensembles, PCA feature rotation slightly impairs single-variable threshold splits.

12 Conclusion

Dimensionality reduction via PCA proved highly effective for linear and distance-based classifiers on the Breast Cancer Wisconsin dataset, retaining top diagnostic performance (Logistic Regression 98.24% CV accuracy, SVM 97.80% CV accuracy) while compressing feature dimensionality from 30 down to 10 components. Stacking and Random Forests maintained robust performance across both spaces, demonstrating the utility of PCA when computational efficiency or multicollinearity reduction is required.

13 References

- [1] Scikit-learn Documentation, “Principal Component Analysis (PCA),” <https://scikit-learn.org/stable/modules/decomposition.html#pca>
- [2] Scikit-learn Documentation, “Supervised Learning Classifiers,” https://scikit-learn.org/stable/supervised_learning.html
- [3] UCI Machine Learning Repository, “Breast Cancer Wisconsin (Diagnostic) Data Set,” <https://archive.ics.uci.edu>